



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

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DNS

❖ DNS

- DNS stands **Domain Name System**.
- DNS is a system that converts/translates a **domain name into an IP address** so that computer can find the correct server on the internet.
OR

The Domain Name System (DNS) is the component of the internet standard protocol responsible for converting human-friendly domain names into the internet protocol (IP) addresses computers use to identify each other on the network.

- Example : www.google.com → **DNS** → 8.8.8.8
- DNS works like a telephone directory / phone book of the internet.
- DNS normally uses port 53 and use UDP.
- DNS is a hierarchical and distributed naming system that translates domain names into IP addresses.
- When you type a domain name like www.google.com into your browser, DNS ensures that the request reaches the correct server by resolving the domain to its corresponding IP address.
- Without DNS users would generally need to remember IP addresses to access internet services.

❖ Domain Name

- A DNS name is a human – readable name assigned to a device, server, or internet service in the DNS system.
- Example : www.google.com , www.amazon.in , www.ppup.ac.in

❖ Structure of DNS / Types of DNS

1. Root

The topmost level of the DNS hierarchy.

- Represented by a dot (.) at the end of a domain name
- Acts as the starting point of domain resolution

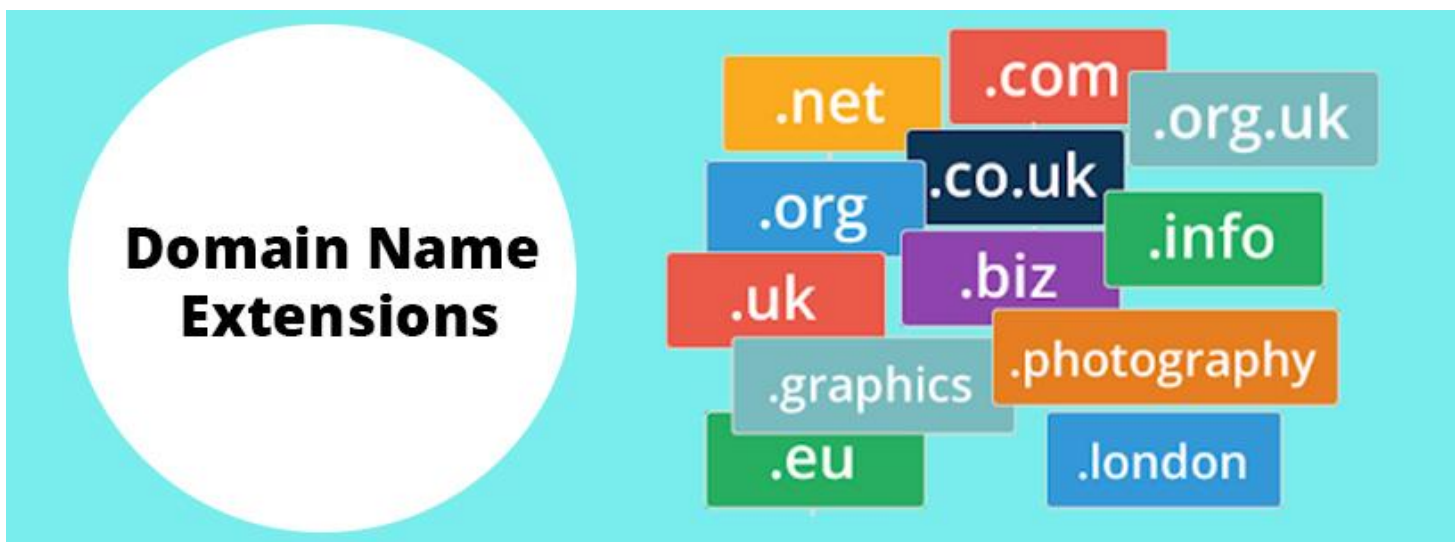
2. Top-Level Domains (TLDs)

The level directly below the root that defines domain extensions.

- Includes extensions like .com, .org, .net, .edu
- Helps categorize domains by purpose or region.
- Example :- .com, .org, .net, .in, .edu,....
 - .com = commercial Organizations
 - .org = Non-profitable organizations
 - .net = Networking organizations
 - .edu = Educational Institutions
 - .gov = Government organizations
 - .mil = Military uses
 - .in = india (all country code – only two letter - provide by ISO)
 - .us = United States (इसे हम ccTLD - Country code TLD कहते हैं)
 - .gov.in = Indian government
 - .gov.us = United state government
 - .bih.in = Bihar
 - .info = information
 - .ac.in = academic institutions in india

Two types of TLD

1. **gTLDs** (Generic Top-Level Domains) → e.g - .com, .net, .org,.....
2. **ccTLDs**(Country code Top-Level Doamins) → e.g - .in, .us, .jp,....



3. Second-Level Domains /registered/main domain

The main domain name registered by an organization.

1. Appears before the TLD (e.g., "example" in example.com)
2. Uniquely identifies a domain under a TLD

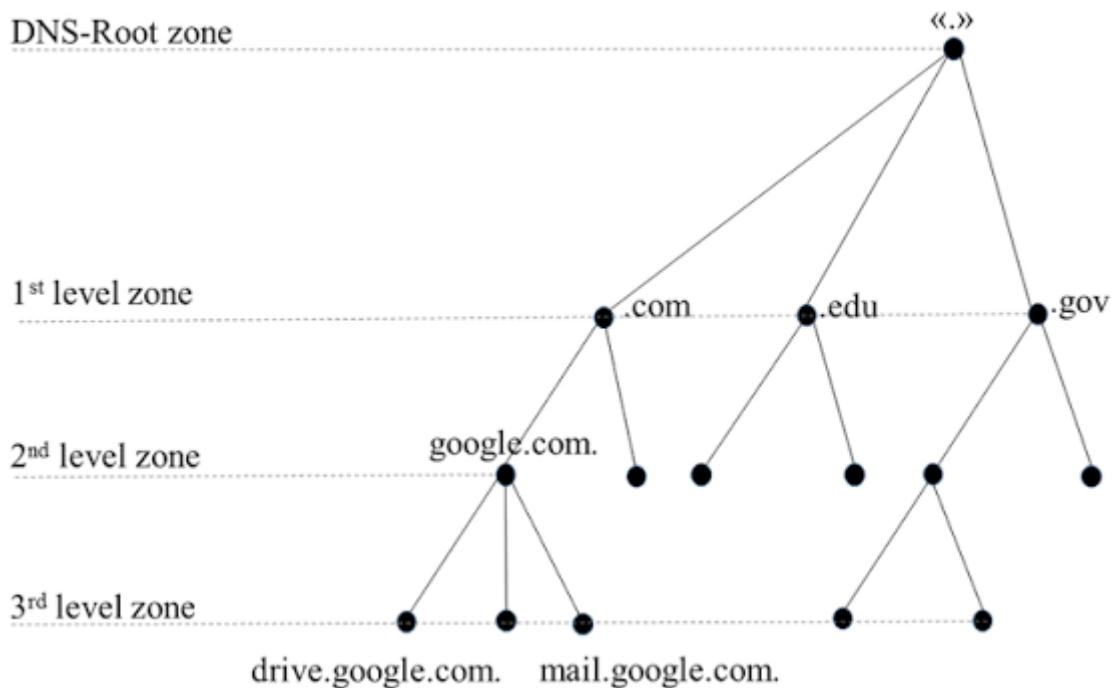
4. Subdomains /host

Extensions of the main domain used for organization.

- Examples: www, mail, blog (mail.google.com → **mail** is subdomains)
- Helps structure different parts of a website

#	Website	TLD	Public Suffix	Registered/Main Domain	Subdomain
1	google.com	.com	.com	google	—
2	youtube.com	.com	.com	youtube	—
3	wikipedia.org	.org	.org	wikipedia	—
4	amazon.in	.in	.in	amazon	—
5	flipkart.com	.com	.com	flipkart	—
6	example.co.in	.in	.co.in	example	—
7	serviceonline.bihar.gov.in	.in	.gov.in	bihar	serviceonline
8	www.bbc.co.uk	.uk	.co.uk	bbc	www
9	www.harvard.edu	.edu	.edu	harvard	www
10	developer.mozilla.org	.org	.org	mozilla	developer

❖ Structure of the DNS Hierarchy



❖ Types of DNS Servers

DNS uses four main types of servers working together to resolve a query:

- (i) **DNS Recruiter / Resolver** : The middleman that receives queries from client applications (like web browsers) and performs the search work.
- (ii) **Root Name Server** : The first step in translating human names to IP addresses; translates root zones (.) and directs to TLD servers. Total 13 logical root server available in world.
- (iii) **TLD Name Server** : Stores location data for top-level domains like .com, .org, or .io.
- (iv) **Authoritative Name Server** : The final stop that holds the actual resource records (IP address mappings) for a domain.

❖ Types of DNS Records

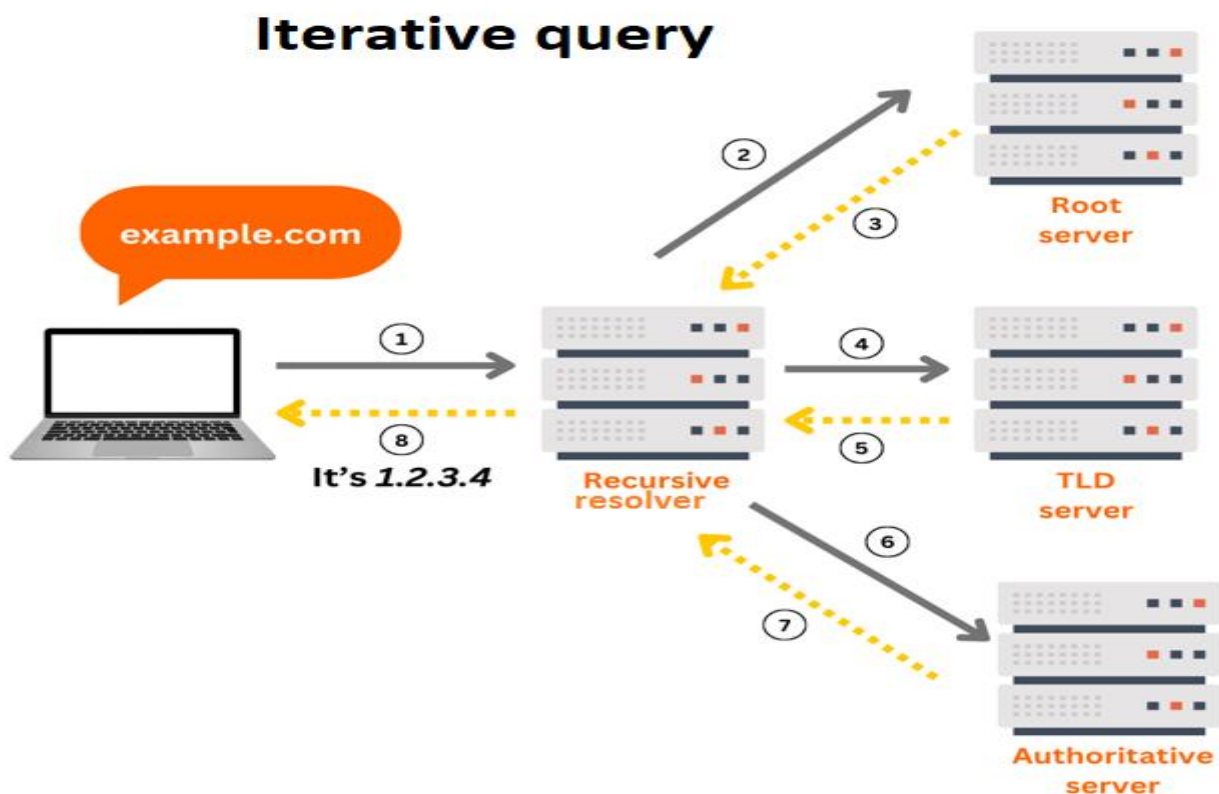
- **A Record (Address)**: Maps a hostname directly to an **IPv4** address (192.0.2.1).
- **AAAA Record (Quad-A)**: Maps a hostname to an **IPv6** address (2001:db8::1).
- **PTR Record (Pointer)**: Resolves an IP address back to a domain name (used in **reverse DNS lookups**).

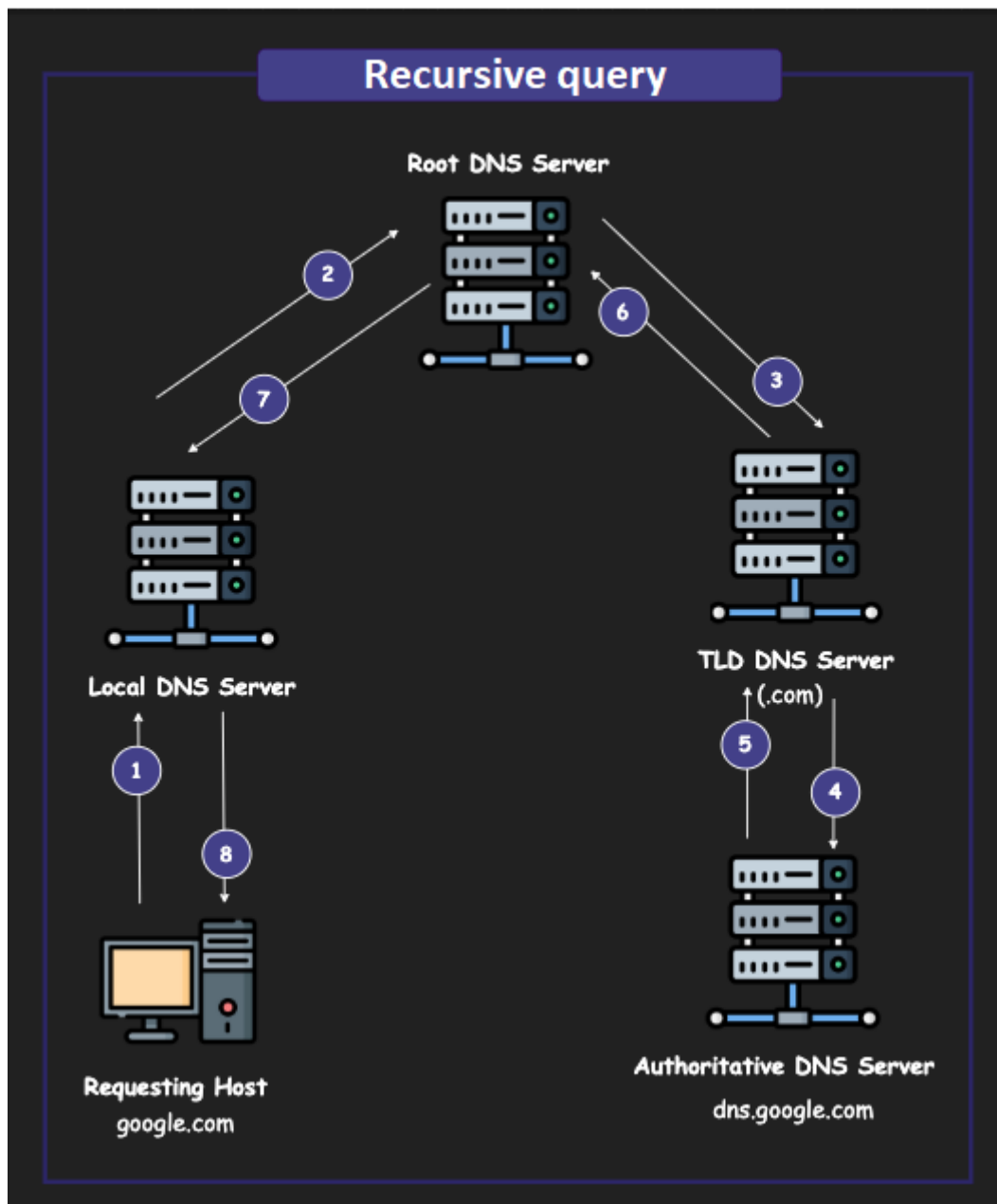
So many.....

❖ Working DNS / Corresponding DNS Server Hierarchy / DNS Query solve

DNS process can be broken down into several steps, ensuring that users can access websites by simply typing a domain name into their browser.

1. **User Input:** The user enters a domain name (e.g. www.example.com) in the browser.
2. **Local Cache Check:** The browser or OS checks its cache for a stored IP address.
3. **DNS Resolver/ Recruiter Query / Recursive resolver/query:** If not found, the request is sent to a DNS resolver (usually by ISP).
4. **Root Server Query:** The resolver queries a root server, which points to the correct TLD server.
5. **TLD Server Response:** The TLD server directs the resolver to the domain's authoritative server.
6. **Authoritative Server Response:** The authoritative server returns the actual IP address.
7. **Final Response:** The resolver sends the IP back to the user, and the browser connects to the server.





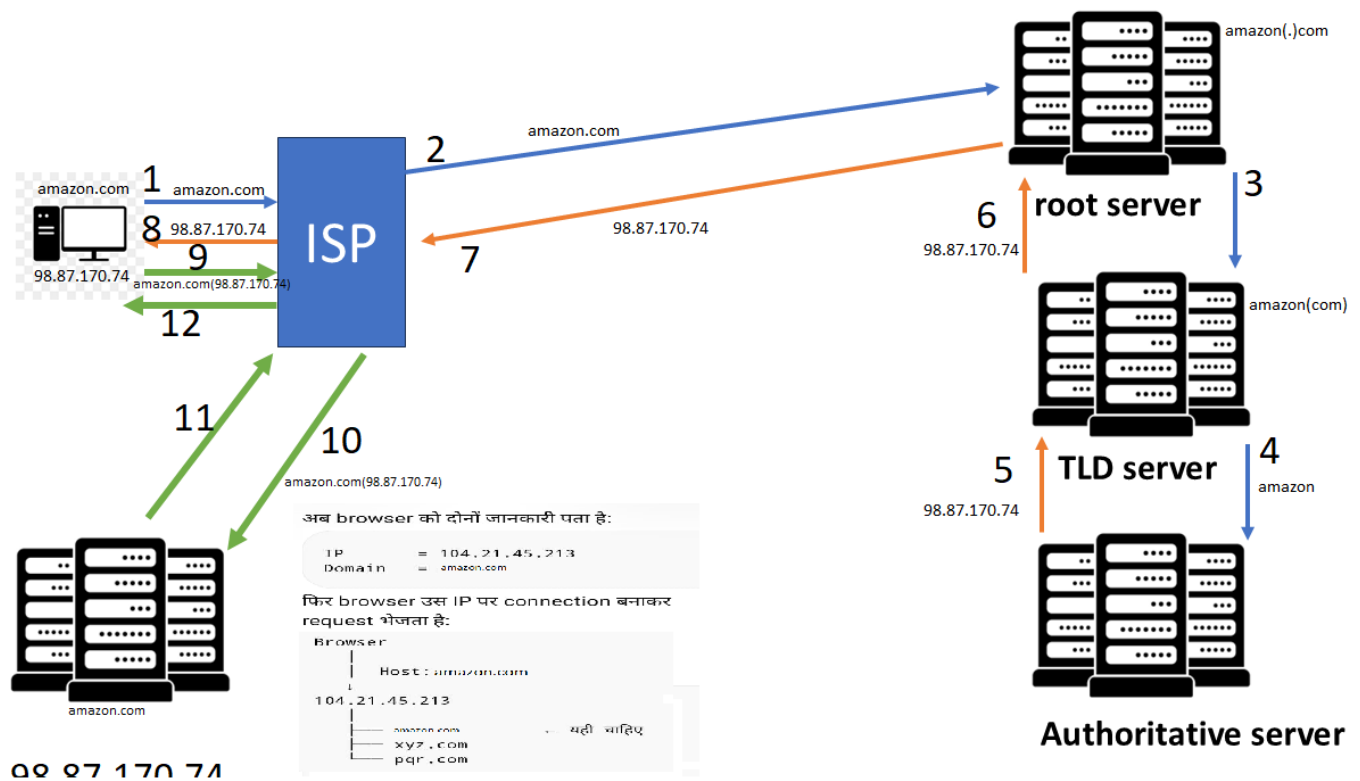
❖ How to see DNS of any websites

- open cmd and type command
- C:\> nslookup website_name
- Example – C:\> nslookup ppup.ac.in

```
C:\Users\hira>nslookup ppup.ac.in
Server: UnKnown
Address: fe80::6e4f:89ff:fee4:8e29

Non-authoritative answer:
Name: ppup.ac.in
Addresses: 2606:4700:3034::ac43:db32
           2606:4700:3030::6815:2dd5
           172.67.219.50
           104.21.45.213

C:\Users\hira>
```



नोट: आप browser में सिर्फ IP address देकर किसी website को नहीं खोल सकते हैं, क्योंकि कोई भी website open होने के लिए browser 2 request server तक ले के जाता है (i) host/domain name (ii) IP address |

जब आप सिर्फ IP देते हैं browser उस server तक तो पहुँच जाता है लेकिन उस server में कौन सा host/domain name चाहिए वो नहीं मिल पाता है, क्योंकि एक server पर एक से ज्यादा website के data रखे होते हैं |

इसलिए DNS बहुत जरूरी हैं |

=====HIRA KUMAR=====